This document explains the other files in this folder, as well as those on GitHub and their contents. At the bottom of this document, there are additional notes and helpful analogies. The GitHub page used is entitled hbll-data_training, and the file is Beginning Python Workshop.

python_workshop_student_download.ipynb

This is the file that we had each attendee download and open in Google Colab. The first half is what instructors will walk through with them. We spent time breaking down each concept, making sure they understood the notation, and knew how they could debug their code. Make sure that they understand the data set before you jump into it. We found it helpful to give them a couple of minutes to walk through the data set without code, just to make sure they knew what it contained.

python_workshop_updated_completed.ipynb

This is the completed version of the practice problems the students go through. It has a solution to the practice problems, as well as notes on what to watch out for. It can be a great resource if you get tripped up on notation for the challenge problems. It is also a great way to gauge how prepared you are for the workshop.

census_and_trends_data_2026-10-7.csv

This is the data that the students need to download. It is helpful to give the attendees a couple of minutes to look through the file, so they understand what data they are working with.

matplotlib_beginner_cheatsheet.png

This is a fun little review sheet with helpful notation the students might need. It has more than what we will cover in this workshop, but it may be helpful when students work on personal projects.

pandas_cheat_sheet.pdf

This is a review sheet with helpful notation that is in the Python Pandas library. It will be a helpful post-workshop review for the attendees.

Additional Notes:

- If you are learning Python for the first time, we recommend LinkedIn Learning courses on Python and Data Science. It's helpful to do a simple Python introduction course before doing one with Data Science.
- O'Reilly Books is something that you have access to with your BYU account. This book is a helpful resource, packed with information: *Python Data Science Handbook, 2nd Edition.* *Preface | Python Data Science Handbook, 2nd Edition*